BCS-079	Unix and Shell Programming	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To enable the students to understand Features of UNIX
CO2	Understanding concept of Open Source Software
CO3	To learn basic commands of Linux
CO4	Understanding about Shell Script
CO5	To Know about Role of system administrator

Detailed Syllabus

MODULE I

Introduction to UNIX: Features of UNIX Operating System,UNIX system organization (the kernel and the shell), Files and directories, Library Functions and system calls, Editors (vi and ed). Introduction to the Concept of Open Source Software, Linux, Linux Architecture, Linux file system (inode, Super block, Mounting and Un- mounting), Essential Linux Commands (grep, fgrep, egrep, make, nmake, gmake, rcs, cvs, sccs, ar, tar, cpio, pax, RPM, autoconfig. Users and permissions- chmod, su, mount, df, fsck, dd, etc), Kernel, Process Management in Linux, Signal Handling, System call, System call for Files, Processes and Signals.

MODULE-II

Programming in shell script: Types of shells, Shell Meta characters, Shell variables, Shell scripts, Shell commands, the environment, Integer arithmetic and string manipulation, Special command line characters, Decision making and loop control, controlling terminal input, trapping signals, arrays. I/O Redirection and Piping, Vi and Emacs editor, Shell control statements, Find, Shell Meta- characters, Shell Scripts, Shell keywords, Shell Procedures and Reporting, Handling documents, scheduling of processes at command, cron, batch commands, Command line argument, Background processes, process synchronization, Sharing of data, user-id, group-id.

MODULE-III

Network Administration: System administration Common administrative tasks, identifying administrative files – configuration and log files, Role of system administrator, Managing user accounts- adding & deleting users, changing permissions and ownerships, Creating and managing groups, modifying group attributes, Temporary disable user's accounts, creating and mounting file system, checking and monitoring system performance file security & permissions, becoming super user using su. **Case study-** Amoeba, Corba, Mac, Eros, Ubuntu.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Ability to remember and understand factual knowledge relevant to system administration tools and technologies
CO2	Ability to write how-to documents, white papers, tutorials guiding other system administrators or users step-by-step through system administration tasks
CO3	Ability to identify abnormal behavior in a computing system, make hypothesis on how to address it, and implement solutions
CO4	Case study of Amoeba, Corba, Mac, Eros, Ubuntu
CO5	Aware about Role of system administrator

Text Book:

1. "Unix Programming Environment" The Kernighan and Pike Prentice – Hall of India

Reference Book:

1. "Unix –Shell Programming" Kochar